

ARR RAFI

Software Engineer – Python/Django & React

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[ARRRAFI10](#) | [in arrafi10](#)

SUMMARY

Software engineer with 1+ year of experience building and operating production Django/React systems at MIST. Delivered platforms for alumni onboarding, graduation registration and payments, and institutional inventory, covering architecture, APIs, databases, frontend delivery, deployment, security, testing, and production support.

TECHNICAL SKILLS

Backend: Python, Django, Django REST Framework, Celery, Django Channels, REST APIs, WebSockets

Frontend: TypeScript, JavaScript, React, Next.js, Vite, Tailwind CSS, responsive web applications

Data: PostgreSQL, Redis, MySQL, schema design, indexing, partitioning, full-text search, query optimization

Cloud & Infrastructure: AWS (EC2, VPC, Lambda, API Gateway), Docker, Linux, Nginx, Gunicorn, PgBouncer, systemd

Quality & Security: pytest, Playwright, GitHub Actions, JWT/OAuth 2.0, RBAC, TOTP 2FA, API security

PROFESSIONAL EXPERIENCE

Military Institute of Science and Technology (MIST), ICT Directorate

Dhaka, Bangladesh

Assistant Programmer

May 2025 – Present

- Sole developer and production maintainer for three institutional platforms, working end to end from requirements and schema design through deployment, monitoring, issue resolution, and user support.
- Served as the Directorate's technical counterpart for externally contracted software, translating requirements, reviewing vendor deliverables, performing acceptance testing, and supporting production handover.
- Built **MISTAS**, a Django/DRF and React alumni platform with 100+ APIs for **1,000+ users**; implemented roster-based identity verification, directory search, JWT/TOTP authentication, and a 68-permission RBAC model. [Live site](#)
- Reworked a Django Channels/Redis messaging bottleneck, reducing measured p95 latency from **32 seconds to approximately 200 ms**; load-tested 1,000 concurrent WebSocket connections.
- Delivered the **MIST 19th Graduation Ceremony Portal** for **1,000+ graduates** using DRF and Next.js; integrated SSLCommerz payment verification, IPN and scheduled reconciliation, QR-coded passes, PDF generation, bulk email, and atomic registration-number issuance. [Live site](#)
- Designed the ICT inventory system around PostgreSQL (46 tables), an append-only audit trail, barcode workflows, 100+ APIs, and an offline scan queue for **2,500+ catalogued products**.
- Established automated backend/frontend testing and GitHub Actions CI; deployed and operated services on self-managed Ubuntu with Nginx, Gunicorn, Celery, Redis, PostgreSQL, and PgBouncer.

SELECTED PROJECTS

QRARG – Research Group and E-Learning Platform

2026

- Built a live React/Django platform with enrollment, lessons, reviews, secure cookie-based JWT, Google OAuth, CSRF protection, rate limiting, and aggregated dashboard APIs; identified and fixed nine authentication and session vulnerabilities through end-to-end testing. [Live site](#)

Job Application Portal

2025

- Developed a Django/React recruitment system supporting approximately 6,000 applicants, SSLCommerz payments, admit cards, roll generation, seat plans, attendance sheets, and applicant administration.

Appointment Management System

2025

- Built a Django/React appointment platform with JWT authentication, scheduling, booking history, prescriptions, email alerts, and role-specific dashboards for administrators, doctors, and patients. [GitHub repository](#)

OrniFire – Aerial Fire Detection with an RC Ornithopter

2024

- Combined a machine-learning fire classifier with an ESP32-based ornithopter for aerial surveillance; the work was subsequently published at IEEE QPAIN 2025.

EDUCATION

B.Sc. in Computer Science and Engineering, MIST

2021 – 2025

CGPA: 3.56/4.00 | Final-year CGPA: 3.89/4.00

RESEARCH & ACHIEVEMENTS

- Co-authored **3 peer-reviewed IEEE conference papers** (2025–2026) covering explainable kidney-disease classification, RC-ornithopter surveillance, and low-light image restoration. [Selected DOI](#)
- Solved **800+ algorithm problems** across LeetCode, Codeforces, and CodeChef; earned 2nd Runner-Up at the Anatolian Rover Challenge 2022 as a member of the MIST Mars Rover Society.
- Completed the **Django Web Framework** professional course by Meta via Coursera.